

The empirical recurrence for OEIS A266020, recovered from its tail

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Abstract

OEIS A266020 records “Empirical recurrence of order 94” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 139$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 96$. Its last coefficient is $c_{94} = -4 \neq 0$, so no further tail satisfies anything shorter; any order-94 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A266020 is “Number of $n \times 3$ integer arrays with each element equal to the number of horizontal, vertical and antidiagonal neighbors exactly one smaller than itself.”. It has offset 1 and begins

$$4, 14, 33, 91, 316, 1045, 3394, 11102, \dots$$

The entry states:

Empirical recurrence of order 94 (see link above)

As of the “Last modified” line on the live entry (Dec 20 2015 07:09:38, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 139$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 139$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 94: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 94.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 278$ exact terms from $n = 2$ on, it returns order 94 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{94} c_i a(n-i),$$

c_1	1	c_{25}	293280	c_{49}	−33373952	c_{73}	1079671
c_2	8	c_{26}	−340525	c_{50}	−100272795	c_{74}	2142342
c_3	−1	c_{27}	−167864	c_{51}	−83656616	c_{75}	2137452
c_4	10	c_{28}	−263019	c_{52}	17864965	c_{76}	798140
c_5	6	c_{29}	808183	c_{53}	101432470	c_{77}	415049
c_6	−113	c_{30}	−88578	c_{54}	76121815	c_{78}	347497
c_7	−5	c_{31}	−5679043	c_{55}	72045497	c_{79}	12509
c_8	−152	c_{32}	6103140	c_{56}	48580962	c_{80}	115494
c_9	−722	c_{33}	7112984	c_{57}	−44591814	c_{81}	16425
c_{10}	164	c_{34}	−11218273	c_{58}	−62891515	c_{82}	−3324
c_{11}	1006	c_{35}	−4124978	c_{59}	−41016131	c_{83}	−34599
c_{12}	2563	c_{36}	5985266	c_{60}	−58303230	c_{84}	−47847
c_{13}	7058	c_{37}	24540203	c_{61}	−30248598	c_{85}	−23981
c_{14}	5833	c_{38}	−4872873	c_{62}	15219689	c_{86}	−13803
c_{15}	−10464	c_{39}	−41964694	c_{63}	38148494	c_{87}	−1646
c_{16}	−20828	c_{40}	22907537	c_{64}	35071696	c_{88}	617
c_{17}	−10517	c_{41}	7290179	c_{65}	13888438	c_{89}	963
c_{18}	−34905	c_{42}	−44207211	c_{66}	14498130	c_{90}	549
c_{19}	19271	c_{43}	−24210974	c_{67}	−3243217	c_{91}	77
c_{20}	63090	c_{44}	36907301	c_{68}	−15684158	c_{92}	4
c_{21}	−94740	c_{45}	110685721	c_{69}	−8761330	c_{93}	−18
c_{22}	90802	c_{46}	2203540	c_{70}	−9244269	c_{94}	−4
c_{23}	102462	c_{47}	−56073940	c_{71}	−5483992		
c_{24}	6582	c_{48}	21048629	c_{72}	−1156429		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 96$, and it is the recurrence OEIS A266020 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{94} - c_1x^{94-1} - \dots - c_{94}$. Its constant term is $-c_{94} = 4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 94 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 94, so $\deg q = 94$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 94, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 94, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -4 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-94 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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